

Basic Principles of Medicine 1

Module: Foundations to Medicine

FTM 50: Nutrition in Health and Disease

Flipped Classroom_After lecture

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Pre-lecture Required reading: For Flipped Class: Nutrition in Health and Disease

- Nutrition: Overview and Macronutrients | Lippincott® Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library (lwwhealthlibrary.com)
<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- Chapter 27 Nutrition: Overview and Macronutrients [27. Nutrition: Overview and Macronutrients | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

Objectives: Flipped Class: Nutrition in Health and Disease

SOM.MK.III.BPM1.1.FTM.3.BCHM.0183	Summarize concept of energy equilibrium: positive and negative energy balance with examples.
SOM.MK.III.BPM1.1.FTM.3.BCHM.0184	Analyze components of daily energy expenditure: a. Resting energy requirement (RMR/BMR) b. Physical activity c. Diet induced thermogenesis.
SOM.MK.III.BPM1.1.FTM.3.BCHM.0185	Indicate macronutrient ratios of a balanced diet. List dietary mono-, di- and polysaccharides.
SOM.MK.III.BPM1.1.FTM.3.BCHM.0186	Describe relationship between dietary lipids and serum cholesterol in healthy individuals.
SOM.MK.III.BPM1.1.FTM.3.BCHM.0187	Highlight significance of macronutrients and micronutrients (vitamins and minerals).
SOM.MK.III.BPM1.1.FTM.3.BCHM.0188	List energy available from macronutrients including ethanol. Calculate energy available from macronutrients if dietary data is provided.
SOM.MK.III.BPM1.1.FTM.3.BCHM.0189	Describe significance of dietary lipids including cholesterol and fatty acids. Identify beneficial effects of supplementation with fish oil: EPA and DHA.
SOM.MK.III.BPM1.1.FTM.3.BCHM.0004	Discuss importance of dietary essential fatty acids: linoleic acid (omega-6) and alpha-linolenic acid (omega-3)
SOM.MK.III.BPM1.1.FTM.3.BCHM.0190	Compare Mediterranean diet and Western diet.
SOM.MK.III.BPM1.1.FTM.3.BCHM.0191	Indicate significance of dietary carbohydrate. Determine significance of glycemic index. Review composition of high fructose corn syrup.
SOM.MK.III.BPM1.1.FTM.3.BCHM.0192	Explore significance of dietary fiber and correlate health benefits of dietary fiber.
SOM.MK.III.BPM1.1.FTM.3.BCHM.0193	Analyze significance of dietary protein and list the dietary essential amino acids. Differentiate between sources of high and low quality protein. Review the significance of combining wheat and beans to improve protein biological value.
SOM.MK.III.BPM1.1.FTM.3.BCHM.0195	Define states of positive and negative nitrogen balance with examples for each. Highlight changes in tissue protein in these states.
SOM.MK.III.BPM1.1.FTM.3.BCHM.0196	Define Respiratory Quotient (RQ). Identify RQ in a mixed diet and during starvation.
SOM.MK.III.BPM1.1.FTM.3.BCHM.0197	Explore roles of ghrelin, cholecystokinin, insulin and leptin involved in appetite regulation.
SOM.MK.III.BPM1.1.FTM.3.BCHM.0199	Describe and review significance of indicators of nutritional status: a. BMI b. Estimation of lean body mass versus adipose tissue. c. Biochemical indicators: Serum albumin

Clinical case for the flipped classroom session

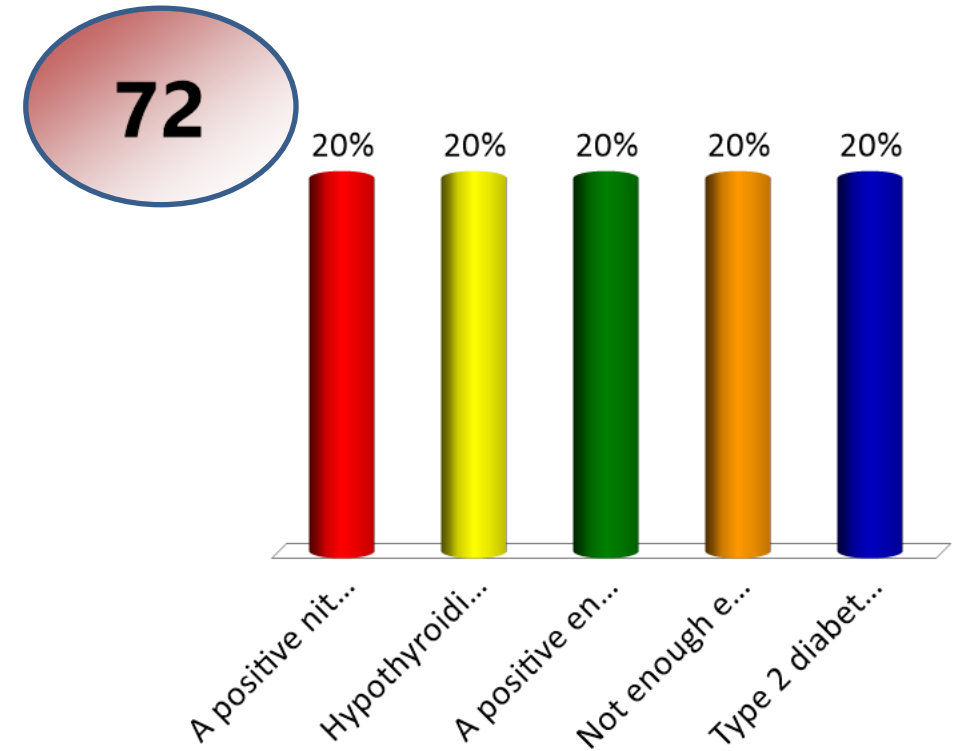
1. A 44-year-old woman consults her physician as she is gaining weight. She describes that she is constantly feeling hungry and experiences uncontrolled food cravings, especially for high-fat and high-sugar food. She tried very hard to lose weight but it did not work.
 - Her height is 5'5" and she weighs 175 pounds and her BMI is 29.2. Her skinfold-thickness is measured and indicates large amounts of subcutaneous fat. Her blood pressure is 125/80 mmHg. Her physician orders a blood laboratory test that includes measurement of Thyroid Stimulating Hormone (TSH) and leptin.
 - Her TSH is 2.0 mU/L (normal for women 0.4 – 2.5 mU/L), her blood glucose is 120 mg/dL (normal 70 - 100 mg/dL) and her leptin level is high with 30 ng/ml (normal 0.5 -15.2 ng/ml) which could indicate leptin resistance as she is borderline obese.
2. To bring her weight and her leptin level down, she agrees to a specific energy restriction diet with a weight loss goal of 10% over the next 6 month. Her diet has a TEE of 1,200 kcal/day with 40% protein, 30% carbohydrate and 30% fat. The protein is fortified with a specific amino acid to stimulate her skeletal muscle synthesis. The diet contains carbohydrates with a low glycemic index and does not contain added sugars. Soluble and insoluble fiber (30 grams/day) is added. The fats contain the dietary essential fatty acids, added fish oil and more oleate than palmitate.
3. After six month she achieves her weight loss goal and her leptin level is in the normal range. She contemplates to switch to the Mediterranean diet with a TEE of 1,800 kcal/day and 20% protein, 42% carbohydrates and 38% fat.

Clinical case for the flipped classroom session

1. A 44-year-old woman consults her physician as she is gaining weight and she is already overweight. She describes that she is constantly feeling hungry and experiences uncontrolled food cravings, especially for high-fat and high-sugar food. She tried very hard to lose weight by exercising but it did not work.
- Her height is 5'5" (1.65 m) and she weighs 175 pounds (97.4 kg). Her skinfold-thickness is measured and indicates large amounts of subcutaneous fat. Her blood pressure is 125/80 mmHg. Her physician orders a blood laboratory test that includes measurement of Thyroid Stimulating Hormone (TSH) and leptin.
- Her TSH is 2.0 mU/L (normal for women 0.4 – 2.5 mU/L), her blood glucose is 120 mg/dL (normal 70 -100 mg/dL) and her leptin level is high with 30 ng/ml (normal 0.5 -15.2 ng/ml) which could indicate leptin resistance as she is borderline obese.

MCQ #1: The 44-year-old woman's weight gain could have been caused by different factors and the physician orders specific tests for his patient. Supported by her data and history, which of the following best explains the weight gain in this patient?

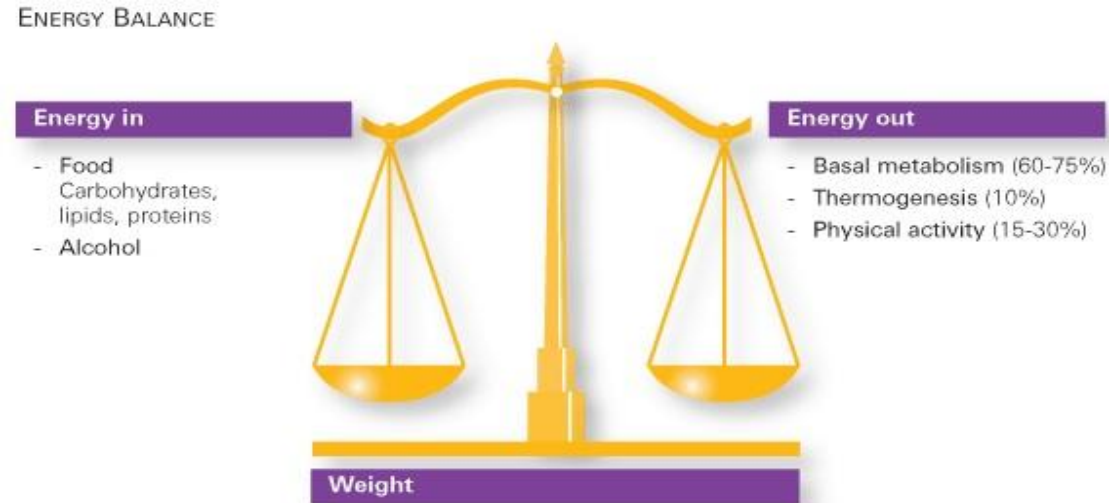
- A. A positive nitrogen balance
- B. Hypothyroidism
- C. A positive energy balance
- D. Not enough exercise
- E. Type 2 diabetes mellitus



MCQ #1: The 44-year-old woman's weight gain could have been caused by different factors and the physician orders specific tests for his patient. Supported by her data and history, which of the following best explains the weight gain in this patient?

- A. A positive nitrogen balance **No** A positive nitrogen balance leads to muscle protein synthesis and **not to subcutaneous fat accumulation. Her diet was rich in fats and sugar.**
- B. Hypothyroidism **No** Hypothyroidism can lead to weight gain but her **thyroid test was normal.**
-  C. A positive energy balance **YES** A positive energy balance means that her **caloric intake exceeds her energy expenditure and weight is gained.**
- D. Not enough exercise **No** exercise may not help her as she has high leptin levels. This could indicate **leptin resistance** where leptin is high but does not lead to the hypothalamus response.
- E. Type 2 diabetes mellitus **No** she has prediabetic blood glucose levels

Healthy individuals are in isocaloric energy balance: weight maintained



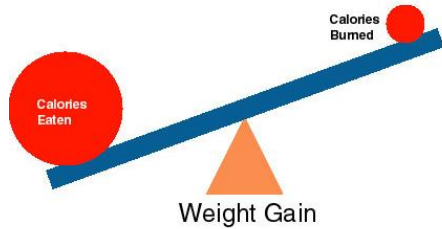
The **Estimated Energy Requirement (EER)** is the average daily dietary energy intake to **maintain** an **energy balance**.

Differences are found in: **genetics, body composition, metabolism, age and gender.**

A positive or negative energy balance can lead to disease

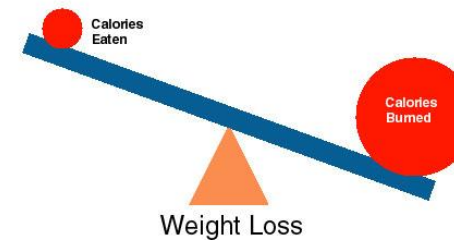
Positive caloric balance

$$\text{In} > \text{Out}$$



Negative caloric balance

$$\text{In} < \text{Out}$$



Weight gain:

- Increased fat stores.
- Can lead to **overweight and obesity**.
- **Higher risk of heart disease and type 2 diabetes mellitus.**

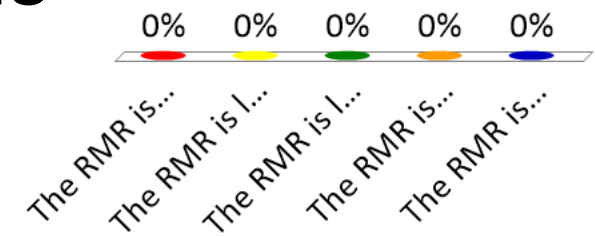
Weight loss:

- Decreased fat stores.
- Excessive weight loss lowers **capacity to fight infections**.
 - Higher risk of **pneumonia or diarrhea**

MCQ #2: The patient's weight gain could be explained by her diet and her resting metabolic rate (RMR). How can the RMR be correctly described and what can change the RMR?

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- A. The RMR is also known as basic metabolic rate
- B. The RMR is lower for fat tissue than for muscle.
- C. The RMR is larger for muscle than for organs
- D. The RMR is about one third of the TEE
- E. The RMR is higher in the elderly



MCQ #2: The patients weight gain could be explained by her diet and her resting metabolic rate (RMR). How can the RMR be correctly described and what can change the RMR?

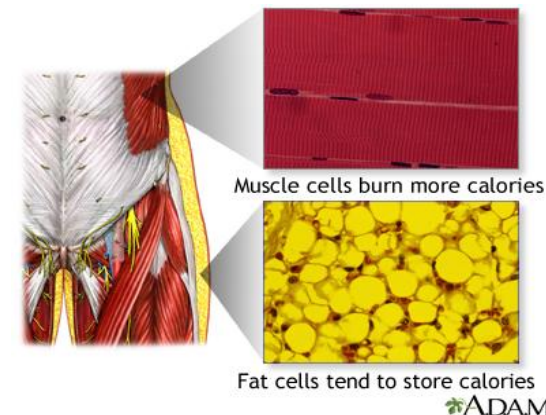
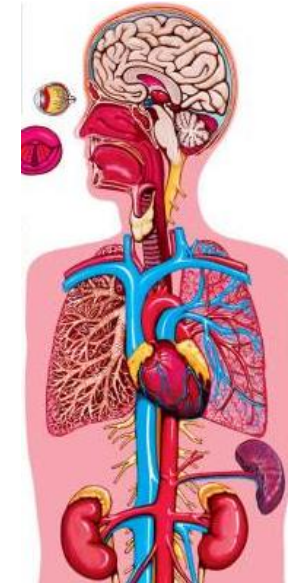
- A. The RMR is also known as BMR **No** the RMR (resting metabolic rate) is about 10% higher than BMR (basic metabolic rate)
-  **B. The RMR is lower for fat tissue than for muscle** **Yes** her skinfold-thickness indicates large amounts of subcutaneous fat leading to a low RMR.
- C. The RMR is larger for muscle than for organs **No** The RMR is larger for organs than for skeletal muscle
- D. The RMR is about one third of the TEE **No** The RMR is about 60 -75 % of the TEE (total energy expenditure per day)
- E. The RMR is higher in the elderly **No** The RMR decreases in the elderly, often resulting from loss of muscle and increase of fat tissue during aging.

Resting metabolic rate (RMR)

- A. Energy expended** in a resting post absorptive state.
- B. Approximately 10% higher** than the basal metabolic rate (**BMR**) which is measured under restrictive conditions.
- C. Affected by** age, gender, body temperature, environmental temperature, thyroid function, pregnancy and lactation.
- D. RMR Increased** in **Hyperthyroidism** (causes weight loss).
RMR Decreased in **Hypothyroidism** (causes weight gain).

Resting metabolic rate (RMR) varies in different tissues

1. **High energy expenditure** is found for the heart, lungs, kidneys, brain and liver.
2. **Organs** have a greater metabolic rate than muscle (15-40 times).
3. **Muscle** has a greater resting metabolic rate than **fat** tissue.



Estimation of daily total energy expenditure (TEE)

BMR is roughly estimated as

24 kcal/kg/day body weight (1 kcal/kg/hour) for males.
(females with less lean muscle mass: 0.9 or 0.8 kcal/kg/hour)

RMR is set in this specific calculation example as 75% of the TEE.

BMR for 60 kg person: $24 \text{ kcal/day} \times 60 = 1,440 \text{ kcal /day}$

RMR = BMR + 10% BMR for this 60 kg person:

$1,440 + 144 \text{ kcal /day} = 1,584 \text{ kcal/day}$

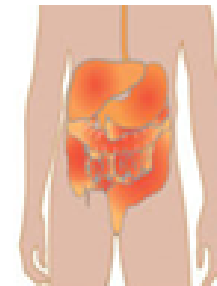
RMR is set as 75% of TEE:

TEE = $1,584 \text{ kcal} / 0.75 = 2,112 \text{ kcal/day}$

Energy output/ expenditure: Energy used per day

Total **energy expenditure (TEE)** per day is the sum of :

1. **Resting metabolic rate (RMR)** to carry out normal **body functions** (60-75%).
2. **Diet-induced thermogenesis (DIT)** (10%).
3. Energy cost of **physical activity** (10-30%).



Respiratory Quotient (RQ) indicates whether mainly carbohydrates or fats are metabolized.

RQ is ratio of oxygen usage to carbon dioxide output.

SUBSTRATE	RQ
Carbohydrate	1.00
Protein	0.84
Fat	0.71

Figure 27.6 p.404

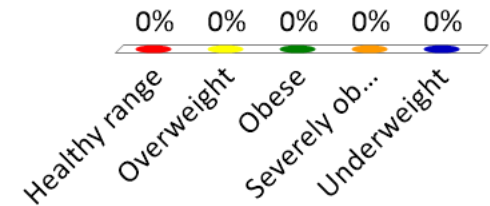
RQ for **carbohydrate metabolism** is **1.0** as for each molecule of oxygen consumed one carbon dioxide molecule is formed. After a meal mostly **carbohydrates** are metabolized and the RQ is 1.

RQ for **fat metabolism** averages **0.7** as for each **100** molecules of **oxygen** only **70 carbon dioxides** are formed due to generation of metabolic water. During starvation mostly **fatty acids** are **metabolized** and the RQ is 0.7.

MCQ #3: The patient's Body mass index (BMI) is 29.2 kg/m². Into which range is her BMI grouped?

72

- A. Healthy range
- B. Overweight
- C. Obese
- D. Severely obese
- E. Underweight



MCQ #3: The patient's Body mass index (BMI) is 29.2.
Into which range is her BMI grouped?

- A. Healthy range **No** the healthy range is between 18.5 to 24.9
- ✓ **B. Overweight** **Yes** she is overweight and borderline to obese
- C. Obese **No** When BMI more than 30
- D. Severely obese **No** When BMI larger than 40
- E. Underweight: When BMI is below 18.5

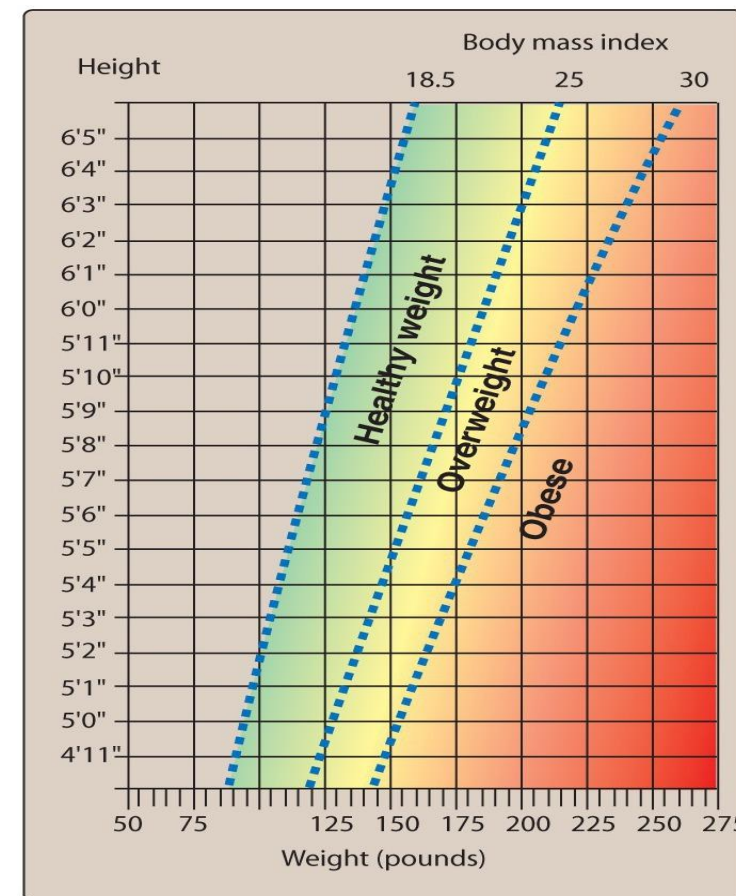
Indicators of nutritional status

Anthropometric measurements

1. Body Mass Index (BMI):

$$(\text{Weight in kg}) / (\text{height in meters})^2$$

- **Underweight** individuals
 - BMI less than 18.5
- **Healthy range**
 - BMI 18.5-24.9
- **Overweight** individuals
 - BMI 25-29.9
- **Obese** individuals
 - BMI greater than 30
 - severe greater than 40



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Figure 26.1 p.390

2. Estimation of body muscle mass versus adipose tissue:



a. **Triceps skinfold (TSF) thickness** estimates subcutaneous **fat stores** (body fat is normally located subcutaneously)

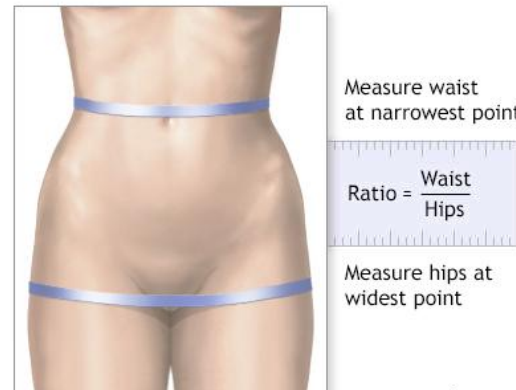
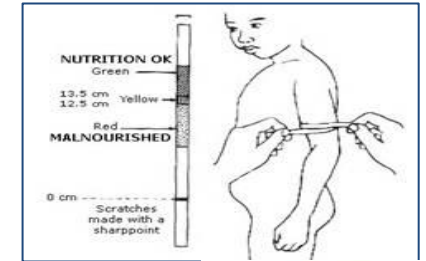
b. **Mid-arm muscle circumference** estimates skeletal **muscle mass**.

c. **Waist circumference** and waist-to-hip ratio: measures visceral fat

c. **Bioelectric Impedance Analysis (BAI)** measures %**body fat**

d. **DEXA scan** (Dual-energy x-ray absorptiometry) measures %**body fat**.

(Also used to measure **bone density**).



SOM.MK.III.BPM1.1.FTM.3.BCHM.0199

c. Biochemical indicators of nutritional status

- Serum albumin level: **Diet deficient in proteins** and amino acids leads to **low serum albumin levels**.
- **Fasting or starvation** leads to an **increase in serum free fatty acids** released from fat cells.

MCQ #4: The laboratory test for the patient's leptin level shows an elevated level which is nearly twice the upper limit of normal. Leptin does not seem to have its effect on her metabolism. Which of the following best describes this hormone?

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- A. Released from the hypothalamus
- B. Leptin and ghrelin stimulate food intake
- C. Leptin level is reduced after a meal
- D. Increases the resting metabolic rate (RMR)
- E. Counteracts insulin action
- F. Higher the adipose tissue, lower the leptin secretion

0% 0% 0% 0% 0% 0%

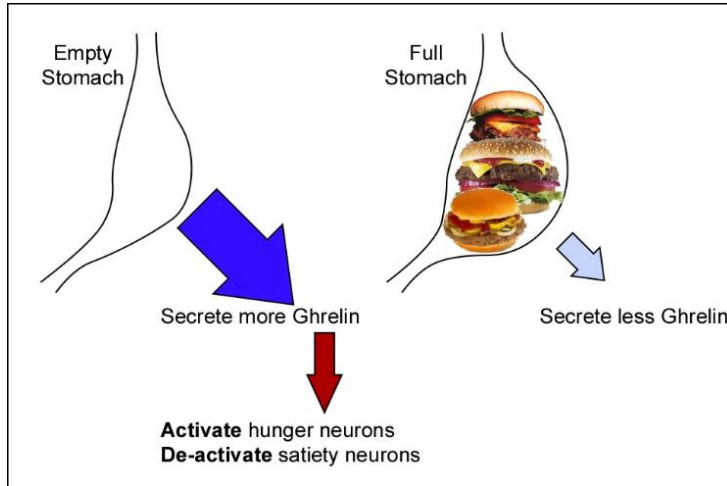
Released fro...
Leptin and G...
Leptin level i...
Increases th...
Counteracts...
Higher the a...

MCQ #4: The laboratory test for the patient's leptin level shows an elevated level which is nearly twice as much as the high normal. But in her case, leptin does not seem to have the normal effect on her metabolism. What are the normal functions of leptin and how is leptin released?

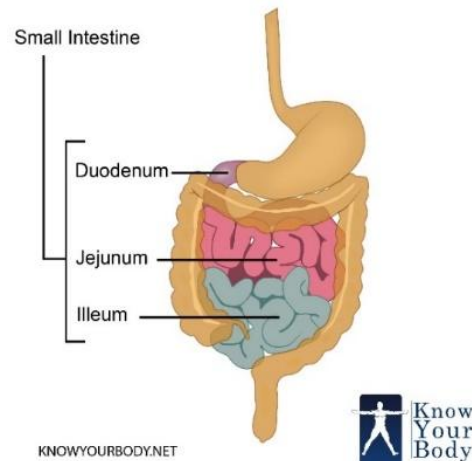
- A. Released from the hypothalamus **No** Leptin is released from **white fat cells** into the blood, passes the blood-brain barrier and has a receptor on the hypothalamus
- B. Leptin and ghrelin stimulate food intake **No** **Leptin** has a satiety effect and **counteracts ghrelin** which signals hunger and food intake
- C. Leptin level is reduced after a meal **No** Leptin level is **increased** after a meal when the fat cells are filled with TAGs
- D.  Increases the resting metabolic rate (RMR) **Yes** **Leptin** regulates the energy balance and weight by stimulating the metabolic rate and increases energy expenditure
- E. Counteracts insulin action **No** **both** leptin and insulin lead to satiety
- F. Higher the adipose tissue, lower the leptin secretion **No** Leptin levels increase with an increase in adipocytes (weight gain)

Appetite regulation: short-time signals

Ghrelin action is counteracted by cholecystokinin

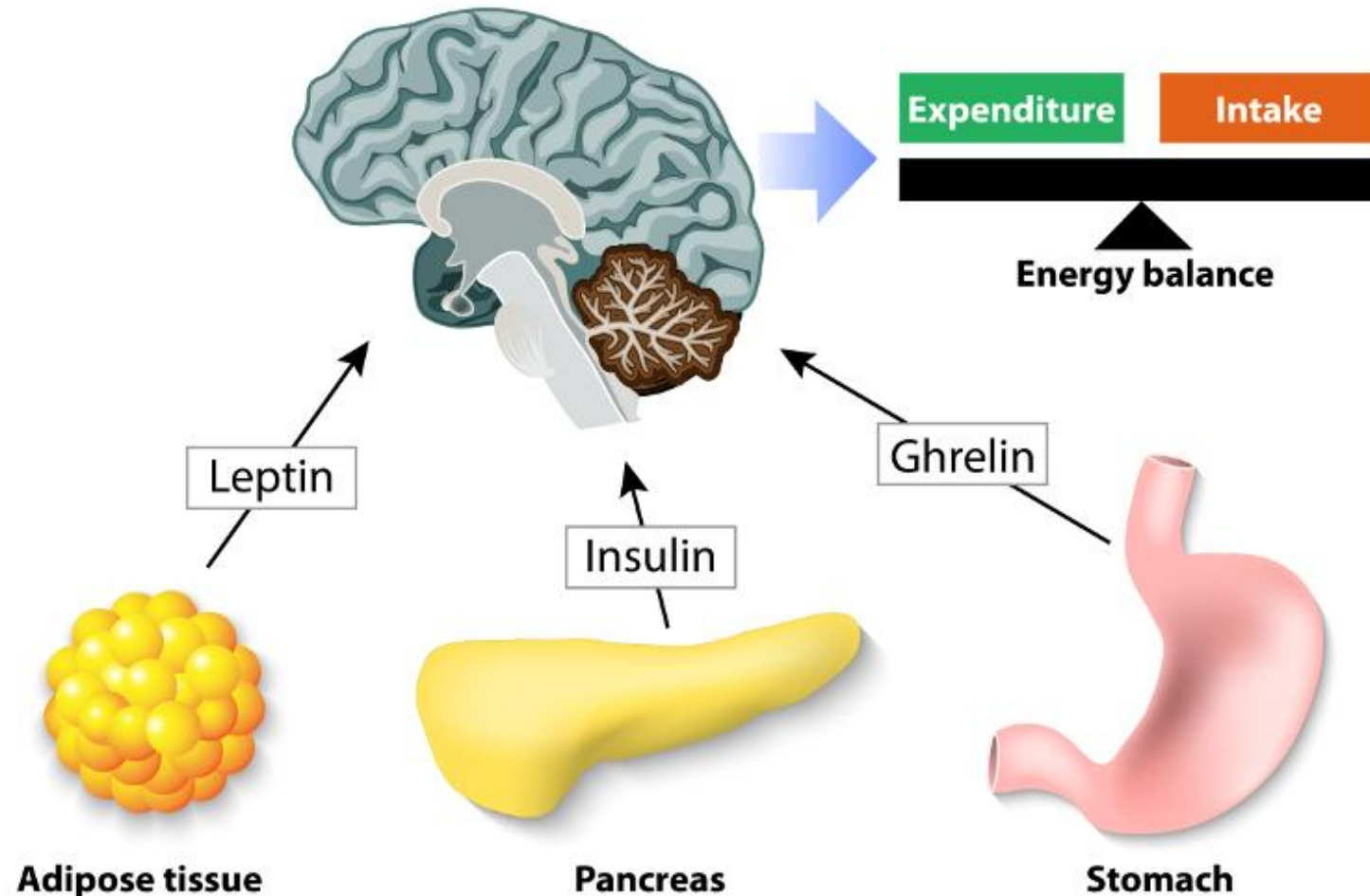


- **Ghrelin: stimulates appetite**
 - Peptide hormone produced by the **stomach**.
 - Stimulates hunger and food intake.



- **Cholecystokinin: stimulates satiety**
 - Peptide hormone produced by the **intestine** (duodenum and jejunum)

Control of food intake and regulation of energy balance



Appetite regulation: Long-term signals

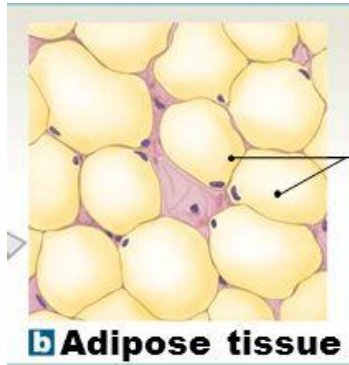
Ghrelin action is counteracted by **leptin** and **insulin**

- **Leptin:** Decreases appetite

- **Peptide hormone** produced by **adipose tissue**.
- **Higher body fat stores → Increases leptin secretion**
- **Stimulates metabolic rate** and **increases energy expenditure**.
- **Overeating** and **chronic high fructose/ refined sugar consumption** may lead to **leptin resistance** and **obesity**.

- **Insulin:** stimulates satiety

- **Peptide hormone** produced by **β -cells** of pancreas.



Clinical case for the flipped classroom session (Continued)

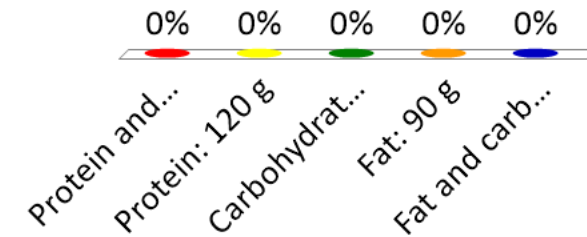
- **2.** To lower her weight and leptin levels, she is prescribed an energy-restriction diet with a weight loss goal of 10% over the next 6 months. Her diet contains 1,200 kcal/day with 40% protein, 30% carbohydrate and 30% fat. The protein is fortified with a specific amino acid to stimulate her skeletal muscle protein synthesis. The diet contains carbohydrates with a low glycemic index and does not contain added sugars. Soluble and insoluble fiber (30 grams/day) is added. The lipids contain the dietary essential fatty acids, added fish oil, and more oleate than palmitate.

MCQ #5: Her diet is energy-restricted with 1,200 kcal/day and has a special composition to reduce her leptin levels.

The macronutrient composition of her diet is: 40% protein, 30% fat, and 30% carbohydrates. Which of the following best describes the macronutrient content of her diet?

- A. Protein and carbohydrates: 40 g each
- B. Protein: 120 g
- C. Carbohydrates: 130 g
- D. Fat: 90 g
- E. Fat and carbohydrates: 130 g each

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MCQ #5: Her diet is energy-restricted with 1,200 kcal/day and has a special composition to reduce her leptin levels.

The macronutrient composition of her diet is: 40% protein, 30% fat, and 30% carbohydrates. Which of the following best describes the macronutrient content of her diet? **You have first to calculate how many kcal/day are needed and then divide by the respective macronutrient kcal/g**

40% protein: $0.4 \times 1,200 = 480$ kcal leads to $480 : 4 = 120$ g of protein

30% carbohydrates: $0.3 \times 1,200 = 360$ kcal leads to $360 : 4 = 90$ g of carbohydrates

30% fat: $0.3 \times 1,200 = 360$ kcal leads to $360 : 9 = 40$ g of fat

- A. Protein and carbohydrates: 40 g each **No** protein 120 g, CHO 90 g
-  **B. Protein: 120 g YES** She needs 480 kcal from protein and that leads 480 divided by 4 kcal/g to 120 g.
- C. Carbohydrates: 130 g **No** she can eat only 90 g of CHO per day
- D. Fat: 90 g: **No** she can only eat 40 g of fat per day
- E. Fat and carbohydrates: 130 g each **No** 40g of fat and 90 g of CHO

Diet provides nutrients necessary for maintenance of health

Macronutrients

Carbohydrates, Fats and Proteins

Provide **energy (calories)** and dietary **essential nutrients**:
Provide essential fatty acids **and** essential amino acids

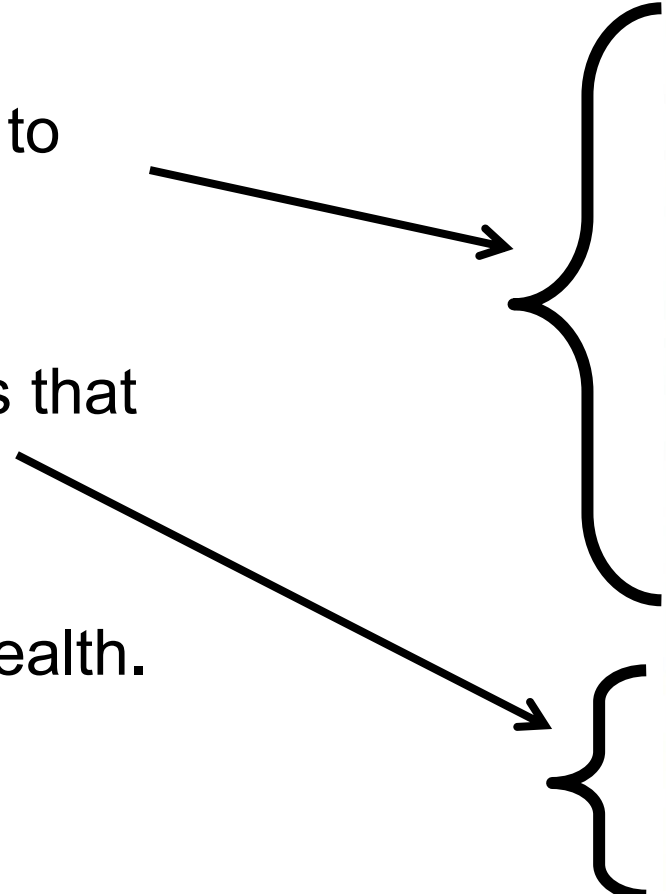
Micronutrients

Vitamins and Minerals:

Needed in **less quantity** and fulfill many functions

Micronutrients: Vitamins

- **Water-soluble** vitamins are mostly used to synthesize coenzymes.
- **Lipid-soluble** vitamins lead to molecules that have different functions:
 - **Vit A**- growth and vision.
 - **Vit D**- calcium metabolism and bone health.
 - **Vit E**- radical scavenging.
 - **Vit K**- blood clotting.

NUTRIENT	
	Thiamine
	Riboflavin
	Niacin
	Vitamin B ₆
	Folate
	Vitamin B ₁₂
	Pantothenic acid
	Biotin
	Choline
	Vitamin C
	Vitamin A
	Vitamin D
	Vitamin E
	Vitamin K

Macronutrient Ratios in balanced diet

associated with lower risk of **chronic disease** and provides **adequate** amounts of **essential nutrients**.

The *Dietary Guidelines for Americans, 2020-2025* is available: <https://www.myplate.gov/>

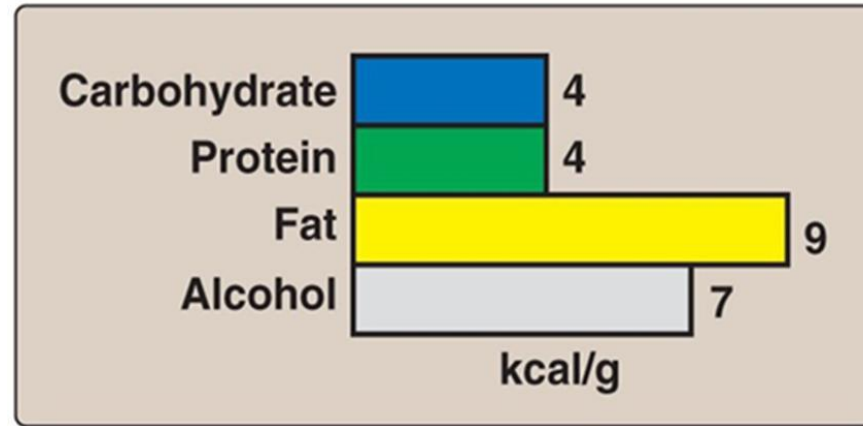
MACRONUTRIENT	RANGE (percent of energy)
Carbohydrate • No less than 130 g/day	45–65
No more than 25% of total calories should come from added sugars.	
Changed to no more than 10%	
Fiber • Men: 38 g • Women: 25 g	
Protein	10–35

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MACRONUTRIENT	RANGE (percent of energy)
Fat	20–35
ω–6 Polyunsaturated fatty acids	5–10
ω–3 Polyunsaturated fatty acids	0.6–1.2*
Approximately 10% of the total fat can come from longer-chain, ω–3 or ω–6 fatty acids.	

● Figure 27.8

Energy from macronutrients and alcohol (kcal/g)



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Figure 27.5 p.403

- **Carbohydrates** and **proteins** generate **4 kcal/g**
- **Fats** generate more than **twice** this amount: **9 kcal/g**.
- **Alcohol** generates nearly the same as **fats**: **7 kcal/g**.
- Note: **Fiber** is **not** calculated to contribute kcal to the diet.

Calculation of macronutrients in a particular diet

Calculation:

For a **diet** with **40% carbohydrate**, **30% fat** and **30% protein**.
TEE assumed as **2,000 kcal/day**.

First: Calculate how many **kcal** of the **respective nutrient** is needed for the **TEE**.
Then **divide** the **kcal** by the respective **kcal/g** to give the amount in **grams**.

Carbohydrate: 40%: $0.4 \times 2,000 \text{ kcal} = 800 \text{ kcal} / 4 \text{ kcal/g} = 200 \text{ g}$

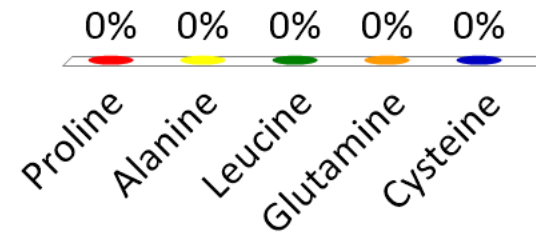
Fats: 30%: $0.3 \times 2,000 \text{ kcal} = 600 \text{ kcal} / 9 \text{ kcal/g} = 67 \text{ g}$

Protein: 30%: $0.3 \times 2,000 \text{ kcal} = 600 \text{ kcal} / 4 \text{ kcal/g} = 150 \text{ g}$

MCQ #6: Which of the following amino acids is dietary essential and is often fortified in the diet in order to stimulate skeletal muscle protein synthesis?

- A. Proline
- B. Alanine
- ✓ C. Leucine
- D. Glutamine
- E. Cysteine

72

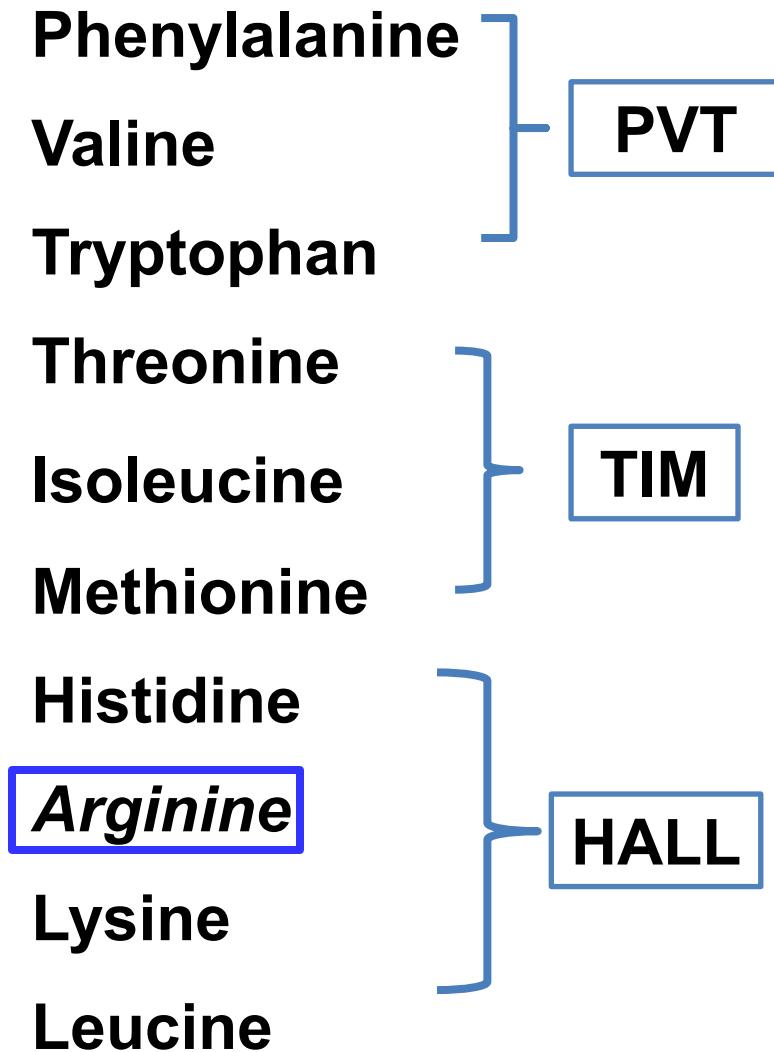


MCQ #6: Which of the following amino acids is dietary essential and is often fortified in the diet in order to stimulate skeletal muscle protein synthesis?

- A. Proline **No** it is not dietary essential
- B. Alanine **No** it is not dietary essential
- ✓ C. **Leucine** **Yes**, this is a dietary essential amino acid and it is known to stimulate skeletal muscle protein synthesis
- D. Glutamine **No** it is not dietary essential
- E. Cysteine **No** it is not dietary essential

Dietary protein provides dietary essential amino acids

Memorize the essential amino acids



Essential amino acids cannot be synthesized in humans and must be provided in the diet

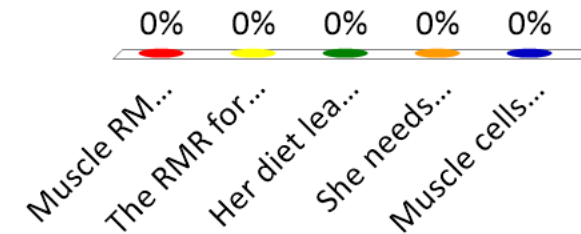
Note: In the past *arginine* was considered as an essential amino acid for adults.

9 amino acids out of 20 amino acids are dietary essential.

MCQ #7: The patient's diet is protein-rich to favor muscle protein synthesis. Why is it of advantage to stimulate her skeletal muscle protein synthesis during her weight loss program?

72

- A. Muscle RMR is higher than organs
- B. The RMR for muscle is higher than for fat
- C. Her diet leads mainly to muscle loss
- D. She needs more muscle for exercising
- E. Muscle cells substitute fat cells



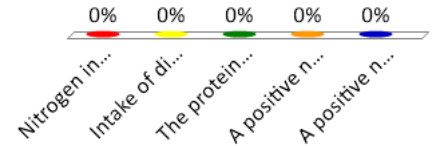
MCQ #7: The patient's diet is protein-rich to favor muscle protein synthesis. Why is it of advantage to stimulate her skeletal muscle synthesis regarding her goal of weight loss?

- A. Muscle RMR is higher than for organs **No** The RMR for organs is higher than for muscle
-  **B. The RMR for muscle is higher than for fat** **YES** her increase of muscle mass would lead to weight loss as the maintenance of muscle uses more energy regarding body temperature
- C. Her diet leads mainly to muscle loss **No** her diet has very high protein % that would prevent muscle loss which can happen in a low carbohydrate diet
- D. She needs more muscle for exercising **No** it is the RMR change that is wanted which would reduce her weight even without exercise
- E. Muscle cells substitute fat cells **No** they cannot substitute fat cells but they have a higher RMR than fat cells

MCQ #8: The patient's diet has 40% protein and her physician follows up by documenting her nitrogen balance. Which of the following best describes nitrogen balance?

72

- A. Nitrogen intake and nitrogen excretion are measured
- B. Intake of dietary essential amino acids and urinary urea are measured
- C. The protein intake and ammonium excretion are measured
- D. A positive nitrogen balance has equal nitrogen intake and excretion
- E. A positive nitrogen balance shows higher urea excretion



MCQ #8: The patient's diet has 40% protein and her physician follows up by documenting her nitrogen balance. What is measured for the nitrogen balance (NB) and what would indicate a positive nitrogen balance?

-  **A. Nitrogen intake and nitrogen excretion are measured YES protein intake and urine urea are measured**
- B. Intake of *dietary essential amino acids* and urea formation are measured **No** the **nitrogen of all amino acids (dietary protein)** is measured for nitrogen intake, but note, for a positive nitrogen balance we need all dietary essential amino acids
- C. The protein intake and *ammonium* excretion are measured **No** the excretion of **urinary urea** is measured
- D. A positive nitrogen balance has equal nitrogen intake and excretion **No** it has higher intake of nitrogen than loss of nitrogen; There is tissue protein synthesis
- E. A positive nitrogen balance shows higher urea excretion **No** that would lead to a negative nitrogen balance

Nitrogen metabolism

Dietary protein requirement can be determined by the
calculating nitrogen balance (NB)

Healthy adults are in **nitrogen balance**

Dietary nitrogen intake
(Protein intake)

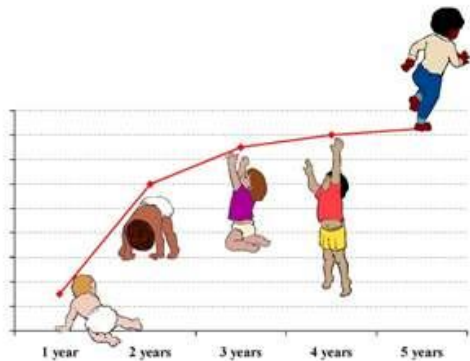


Nitrogen Loss
(Urinary urea nitrogen
+ loss in sweat ...)

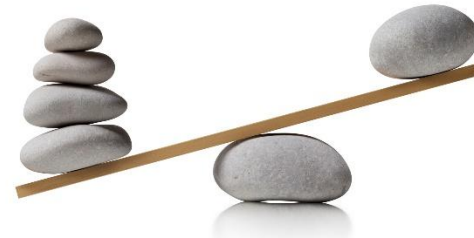
Nitrogen intake equals nitrogen loss

Positive nitrogen balance

Dietary amino acids are used for tissue **protein synthesis** and **less urea** is formed.



High dietary nitrogen
intake (Protein)



Low nitrogen loss
(Urine Urea)

Nitrogen intake exceeds nitrogen excretion.

Characterized by tissue growth

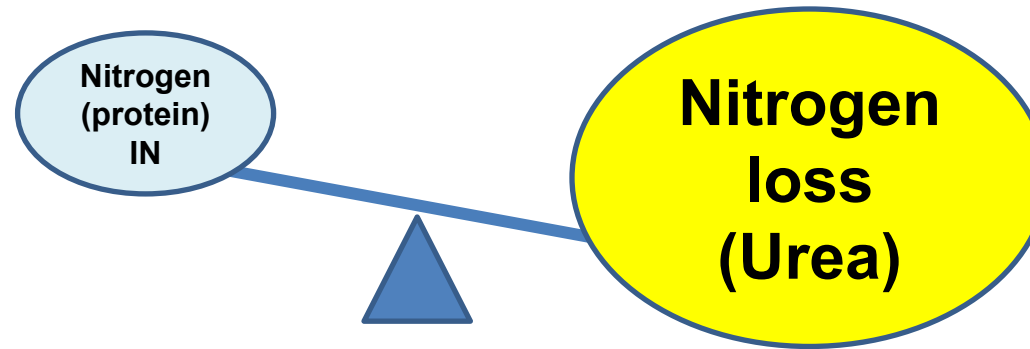
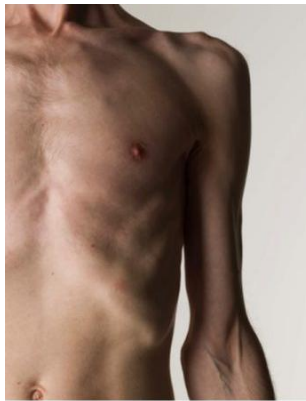
Observed in infants, children, adolescents, pregnant females or during recovery from an emaciating illness.

Note: Provide **all essential** amino acids in diet

Negative nitrogen balance

Dietary nitrogen intake is less than nitrogen loss.

Less protein synthesis and more **muscle protein degradation**.



Nitrogen loss exceeds nitrogen intake.

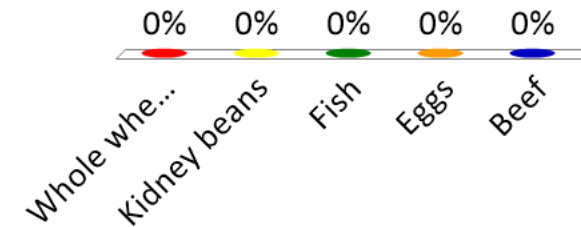
Observed in:

- a. Inadequate dietary protein** and lack of dietary **essential amino acids**.
- b. Stress:** trauma, burns, illness or surgery
- c. Elderly**

MCQ #9: Food protein is evaluated related to the content of dietary essential amino acids. Which of the following dietary sources has all dietary essential amino acids and has a high amino acid score?

- A. Whole wheat bread
- B. Kidney beans
- C. Fish
- D. Eggs
- E. Beef

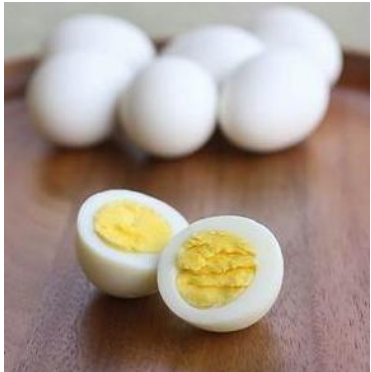
72



MCQ #9: Food protein is evaluated related to the content of dietary essential amino acids. Which food shown below has all dietary essential amino acids and has a high amino acid score?

- A. Whole wheat bread
- B. Kidney beans
- C. Fish
- ✓ **D. Eggs** YES eggs and milk and soy beans contain all dietary essential amino acid with a score of 1.00 for the protein digestibility-corrected amino acid score.
- E. Beef

Relative quality of common dietary proteins (essential amino acids)



Source	PDCAAS value
Animal proteins	
Egg	1.00
Milk protein	1.00
Beef/poultry/fish	0.82–0.92
Gelatin	0.08
Plant proteins	
Soybean protein	1.00
Kidney beans	0.68
Whole wheat bread	0.40

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Figure 27.17 p. 412



PDCAAS = **p**rotein **d**igestibility **c**orrected **a**mino **a**cid **s**core.

Score based on **profile** of **essential amino acids** and **digestibility**

Low score indicates essential amino acid deficiency or poor digestibility

Proteins from plant sources

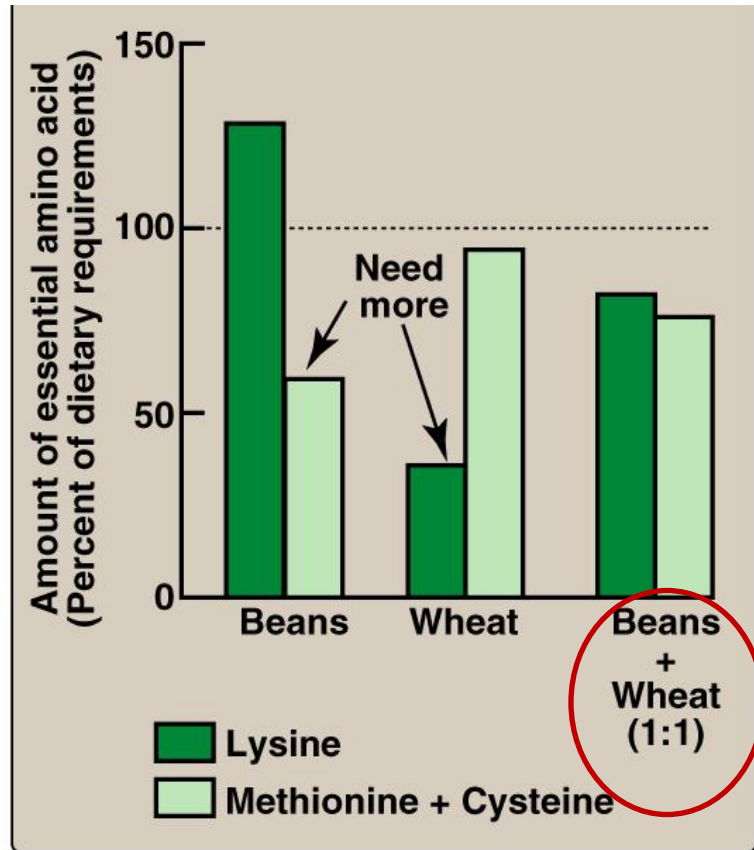


Figure 27.18 p.412

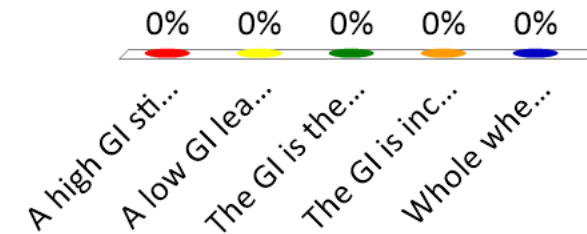
Some plant protein sources may have lower amounts of an essential amino acid. A mixture of plant protein (beans+ wheat) sources optimizes essential amino acid content



MCQ #10: The patient's diet is designed for weight loss and carbohydrates with low glycemic index (GI) are included. Which of the following best describes this index?

- A. A high GI stimulates the release of ghrelin
- B. A low GI leads to more insulin release
- C. The GI is the area under the blood glucose curve
- D. The GI is increased by dietary fiber
- E. Whole wheat bread has a higher GI than white bread

72



MCQ #10: The patient's diet is designed for weight loss and in addition carbohydrates with low glycemic index (GI) are chosen. What is described by the GI and what results from a low or high GI?

- A. A high GI stimulates the release of ghrelin **No** ghrelin release is stimulated by an empty stomach
- B. A low GI leads to more insulin release **No** it leads to less insulin fluctuation
- ✓ **C. The GI is the area under the blood glucose curve** **Yes** and glucose is often the control. In the past white bread was used as control, you have to watch out for that, then the numbers change.
- D. The GI is increased by fiber **No** it is decreased by fiber
- E. Whole wheat bread has a higher GI than white bread **No** the other way round, whole wheat bread has a lower GI; Presence of dietary fiber lowers the GI in whole wheat bread



Dietary carbohydrates

(45 - 65% of energy)



- **Monosaccharides:**
 - Mainly **glucose** and **fructose**.
 - **Fructose** is perceived **sweeter** than glucose.
 - Abundant in fruits and honey.
- **Disaccharides:**
 - **Sucrose:** Glucose + **fructose** (table sugar)
 - **Lactose:** Glucose + **galactose** (milk sugar)
 - **Maltose:** Glucose + **glucose** (malt sugar)
- **Polysaccharides:**
 - **Starch:** found in abundance in **plants**.
 - **Glycogen:** found in **animals**.

Dietary carbohydrates and health concerns

Diets rich in refined/added **sugar** can increase risk of **obesity, diabetes** and **caries**



High-fructose corn syrup (HFCS 55) (55% fructose and 42% glucose).

Contains a mixture of monosaccharides

Consumption of high levels of fructose may be harmful.

Foods with **high glycemic index** sharply **elevate** blood **glucose** levels resulting in **higher serum insulin levels**.

Glycemic index (GI)

1. **Glycemic index** defined as **area under the positive blood glucose response curve** when **compared** to control. (**control** is 50 grams of **glucose** which is set at **100**)

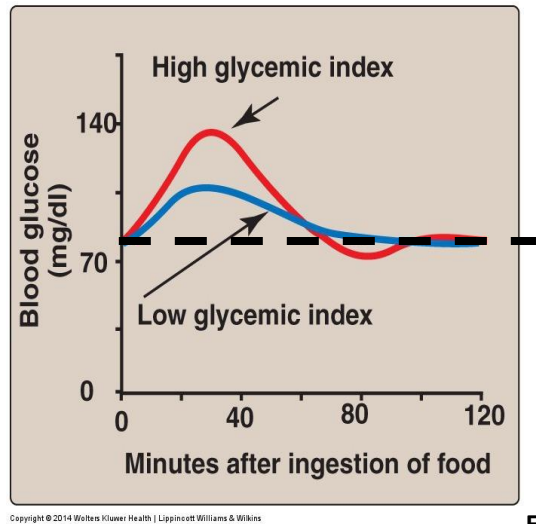
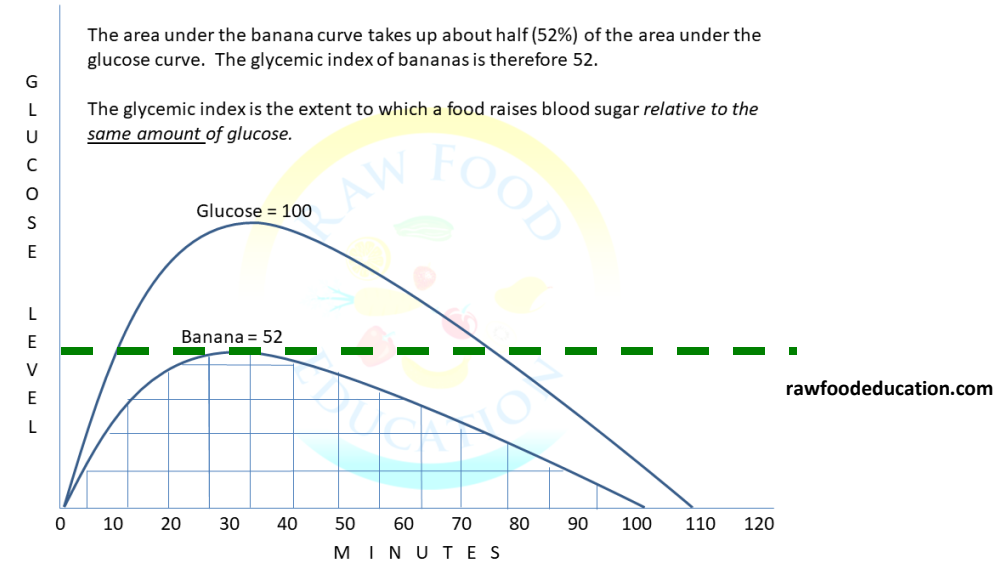


Figure 27.16 p.411



2. Bananas (GI=52) because area under the banana curve is 52% of that of glucose.
3. **GI affected** by several dietary factors like **carbohydrate digestibility**, **fiber**, and **fat content**.

Benefits of low glycemic index

High GI: above 70.

Low GI: less than 55.



Low glycemic index diet is beneficial:

(whole wheat, vegetables, legumes, fruit)

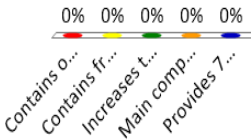
- Slowly **increase** blood **glucose**.
- **Sustain** blood **glucose** levels longer.
- Cause **less fluctuation** of **insulin** release.
- **Improve** blood **glucose** control in **diabetic patients**.
- **Increase satiety** and may help to **limit caloric** intake.



MCQ #11: The patient's diet is low in carbohydrates and does not contain added sugars like high-fructose corn syrup (HFCS). Which of the following best describes this additive?

72

- A. Contains only fructose
- B. Contains fructose and sucrose
- C. Increases the glycemic index
- D. Main component of honey
- E. Provides 7 kcal/g



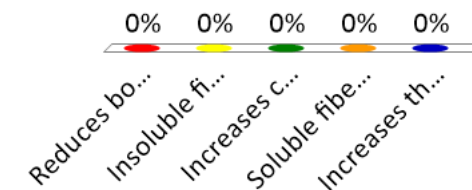
MCQ #11: The patient's diet is low in carbohydrates and does not contain added sugars like glucose or high- fructose corn syrup (HFCS). How can you describe HFCS 55?

- A. Contains only fructose **No** Contains fructose (55%) and glucose
- B. Contains fructose and sucrose **No** it contains only monosaccharides; Sucrose is a disaccharide
-  C. **Increases the glycemic index** **Yes** HFCS contributes to added sugars and is rapidly absorbed
- D. Main component of honey **No** honey is a product of bees and it contains natural fructose
- E. It provides 7 kcal/g **No** it provides like all carbohydrates 4 kcal/g

MCQ #12: Many weight-loss diets are fortified with soluble and insoluble fiber. Which of the following best describes this dietary constituent?

72

- A. Reduces bowel motility
- B. Insoluble fiber is partially digested by bacteria
- C. Increases cholesterol absorption
- D. Soluble fiber can lead to feeling of satiety
- E. Increases the rate of glucose absorption



MCQ #12: Many weight-loss diets are fortified with soluble and insoluble fiber. How do they act differently and what is the overall advantage of all fiber?

- A. Dietary fiber reduces bowel motility **No** it increases bowel motility and **prevents constipation**; The advantage of fiber is that it softens stool by drawing water into the lumen of the intestine and prevents constipation and hemorrhoids.
- B. Insoluble fiber is partially digested by bacteria. **No** soluble fiber is partially digested by bacteria in the large intestine; Can give rise to flatulence when excess fiber in diet
- C. Increases cholesterol absorption **No** Fiber decreases cholesterol absorption
-  **D. Soluble fiber can lead to feeling of satiety Yes and that can prevent overeating; Fiber does not contribute to dietary calories**
- E. Dietary fiber increases the rate of glucose absorption **No** The rate of glucose absorption is decreased by fiber. Presence of fiber lowers the glycemic index. Fiber also reduces absorption of dietary cholesterol and facilitates loss of bile acids in stools

Health benefits of Dietary fiber

Fiber can absorb 10-15 times its own weight in water.

Drawing water into the lumen of the intestine increases bowel motility and movement.

Soluble fiber fermented by gut bacteria whereas **insoluble** fiber passes through the digestive tract largely intact.

Soluble fiber can be found in foods such as oatbran, barley, nuts, seeds, beans, lentils, fruits (citrus, apples), strawberries and many vegetables



Soluble fiber sources



Insoluble fiber is found in foods such as whole wheat and whole grain products, vegetables, and wheat bran

Insoluble fiber sources



ADAM

Health effects

Reduces constipation and hemorrhoid formation, softens stools

Increases bowel motility, thus reducing exposure of gut to carcinogens

Decreases absorption of dietary fat and cholesterol
Increases fecal loss of cholesterol

Delays gastric emptying
Generates sensation of fullness
Reduces postprandial blood glucose concentration

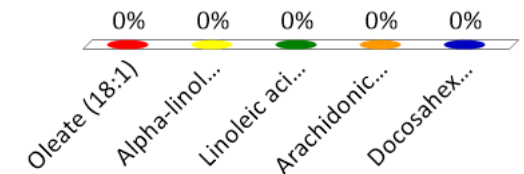


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MCQ #13: The patient's weight reduction diet contains 30% fat which is less than that in the Mediterranean diet (38%). Her diet contains the two dietary essential fatty acids, fish oil and oleate. Which fatty acid shown is a dietary essential fatty acid of the omega-3 family?

- A. Oleate (18:1)
- B. Alpha-linolenic acid (18:3)
- C. Linoleic acid (18:2)
- D. Arachidonic acid (20:4)
- E. Docosahexaenoic acid (22:6)

72



MCQ #13: The patient's weight reduction diet contains 30% fat which is less than the 38% used in the Mediterranean diet. Her diet contains the two dietary essential fatty acids, fish oil and oleate. Which fatty acid shown is a dietary essential fatty acid of the omega-3 family?

- A. Oleate (18:1) **No** oleic acid is not dietary essential
- ✓ **B. Alpha-linolenic acid (18:3) Yes it is an omega-3 dietary essential fatty acid**
- C. Linoleic acid (18:2) **No** it is dietary essential but of the omega-6 family
- D. Arachidonic acid (20:4) **No** it can be formed from linoleic acid and is also of the omega-6 family
- E. Docosahexaenoic acid (22:6) **No** it is an omega-3 fatty acid but not dietary essential, as it can be formed from linolenic acid via eicosapentaenoic acid (EPA). Please note that DHA is especially needed for brain development and function and for vision.



Dietary lipids

(20 - 35% of energy)

Total Fat	12g
Saturated Fat	3g
Trans Fat	0g
Cholesterol	20mg
Sodium	470mg

- Provide **energy for metabolism** and **storage** of **fat** in adipose tissue.
- **Triacylglycerides (TAG)** represent most of the dietary **fat**
- **Fatty acid composition of TAG** is significant for health
 - **Low saturated fatty acids and AVOID trans-fatty acids.**
 - **Monounsaturated and polyunsaturated** fatty acids are beneficial.
 - **Dietary essential fatty acids** needed for membrane fluidity, are precursors for **eicosanoids** and can cross the blood-brain barrier.
- **Essential fatty acid deficiency can lead to scaly dermatitis.**
- Diet must provide **lipid-soluble vitamins.**

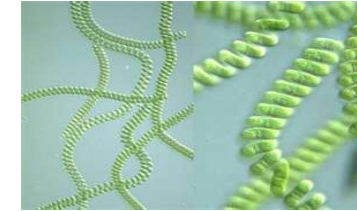
Dietary essential fatty acids and their products



Linoleic acid 18:2, ω 6



Arachidonic acid 20:4, ω 6
Eicosanoids



α -Linolenic acid 18:3, ω 3



EPA 20:5, ω 3
Eicosanoids



DHA 22:6, ω 3
Brain, vision



Dietary omega-3 fatty acids and cholesterol



- Supplementation with fish oil provides omega-3 fatty acids EPA and DHA which improve **brain** function and **vision**.
- **Physiological beneficial** effects of EPA and DHA include reduction of
 - Heart rate
 - Arrhythmia
 - Blood pressure
- **Cholesterol** component of **plasma membranes**; and for **synthesis** of **bile acids**, **steroid hormones**, and **vitamin D**.
 - If not provided by the diet, cholesterol is synthesized in the liver (not essential in diet)



AHA Recommended dietary cholesterol intake

Healthy individuals: less than 300 mg/day.

People with **diabetes** or **heart disease**: less than 200 mg/day

- **Dietary cholesterol** has **less influence** on fasting plasma LDL cholesterol levels in **healthy** individuals
- Higher dietary **saturated fatty acids (palmitate)**, and **trans-fatty acids** lead to high serum cholesterol and **increased risk of CHD**.
- Limiting animal fats, may reduce intake of saturated fatty acids.
- One egg yolk contains around 185 mg cholesterol. Eggs have a perfect score for dietary essential amino acids and contain many essential vitamins and minerals.

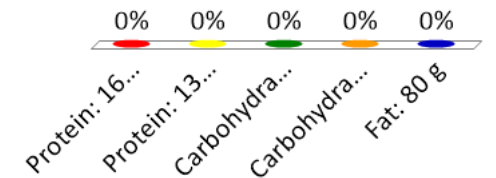
Clinical case for the flipped classroom session (Continued)

- **3.** After six months, she achieves her weight loss goal, and her leptin level is in the normal range. She contemplates switching to the Mediterranean diet with a TEE of 1,800 kcal/day and 20% protein, 42% carbohydrates and 38% fat.

MCQ #14: If the patient would switch to a diet with a TEE of 1,600 kcal/day and 30% protein, 40% carbohydrate and 30% fat, how many grams of these macronutrients can she eat per day ?

- A. Protein: 160 g
- B. Protein: 130 g
- C. Carbohydrate: 120 g
- D. Carbohydrate: 160 g
- E. Fat: 80 g

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MCQ #14: If she would switch to a diet with a TEE of 1,600 kcal/day and 30% protein, 40% carbohydrate and 30% fat, how many grams of these macronutrients can she eat per day ?

Again, you calculate how many kcal per macronutrient, and then divide by 4 for protein and CHO and by 9 for fat.

Protein: $0.3 \times 1,600 = 480$ kcal divided by 4 leads to 120 g of protein

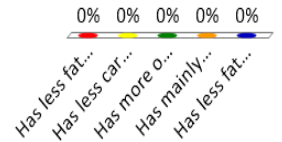
CHO: $0.4 \times 1,600 = 640$ kcal divided by 4 leads to 160 g of carbohydrate

Fat: $0.3 \times 1,600 = 480$ kcal divided by 9 leads to 53 g of fat

- A. Protein: 160 g **No** only 120 g of protein
- B. Protein: 130 g **No** only 120 g of protein
- C. Carbohydrate: 120 g **No** she can eat 160 g of CHO
- ✓ **D. Carbohydrate: 160 g YES she would be able to eat 160 g of carbohydrates**
- E. Fat: 80 g **No** only 53 g of fat

MCQ #15: The Typical Mediterranean diet is compared to the Typical Western diet. Which of the following best describes the Mediterranean diet?

- A. Has less fat and more carbohydrates
- B. Has less carbohydrates and more protein
- C. Has more oleic acid and less palmitic acid
- D. Has mainly omega-3 fatty acids
- E. Has less fats than the Western diet



72

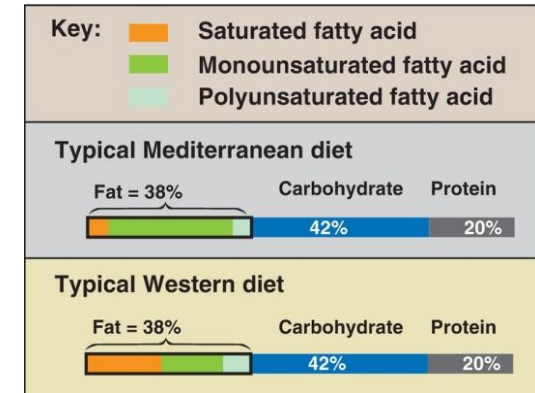
MCQ #15: The Typical Mediterranean diet was compared to the Typical Western diet. What is special for the Mediterranean diet?

- A. Has less fat and more carbohydrates **No** they have the same % fat and same% CHO
- B. Has less carbohydrates and more protein **No** they have the same % CHO and same % protein
-  **C. Has more oleic acid and less palmitic acid** **Yes** this might lead to prevention of CHD, together with radical scavengers found in virgin olive oil
- D. Has mainly omega-3 fatty acids **No** it has mainly oleic acid from virgin olive oil
- E. Has less fat than the Western diet **No** the % fat is the same but the fatty acid composition is different. **The beneficial effect is mainly related to the fatty acid composition and virgin olive oil.**

Mediterranean diet compared to the Western diet

Both diets have **38% of fat** but different composition.

Mediterranean diet has **more** monounsaturated fatty acids (**oleic acid 18:1**) and **less** saturated fats (**palmitate**).



Mediterranean diet lowers lipid cardiovascular risk factors



↑ Olive oil (extra-virgin)
Fish oils, nuts (ω 3 fatty acids)
Plant-based food, fiber

↓ Low Saturated fat
Low Red meat



**Thank you,
please email if you have
questions.
All the best for your exam!**

Dr. Mohamed Idries
midries@sgu.edu